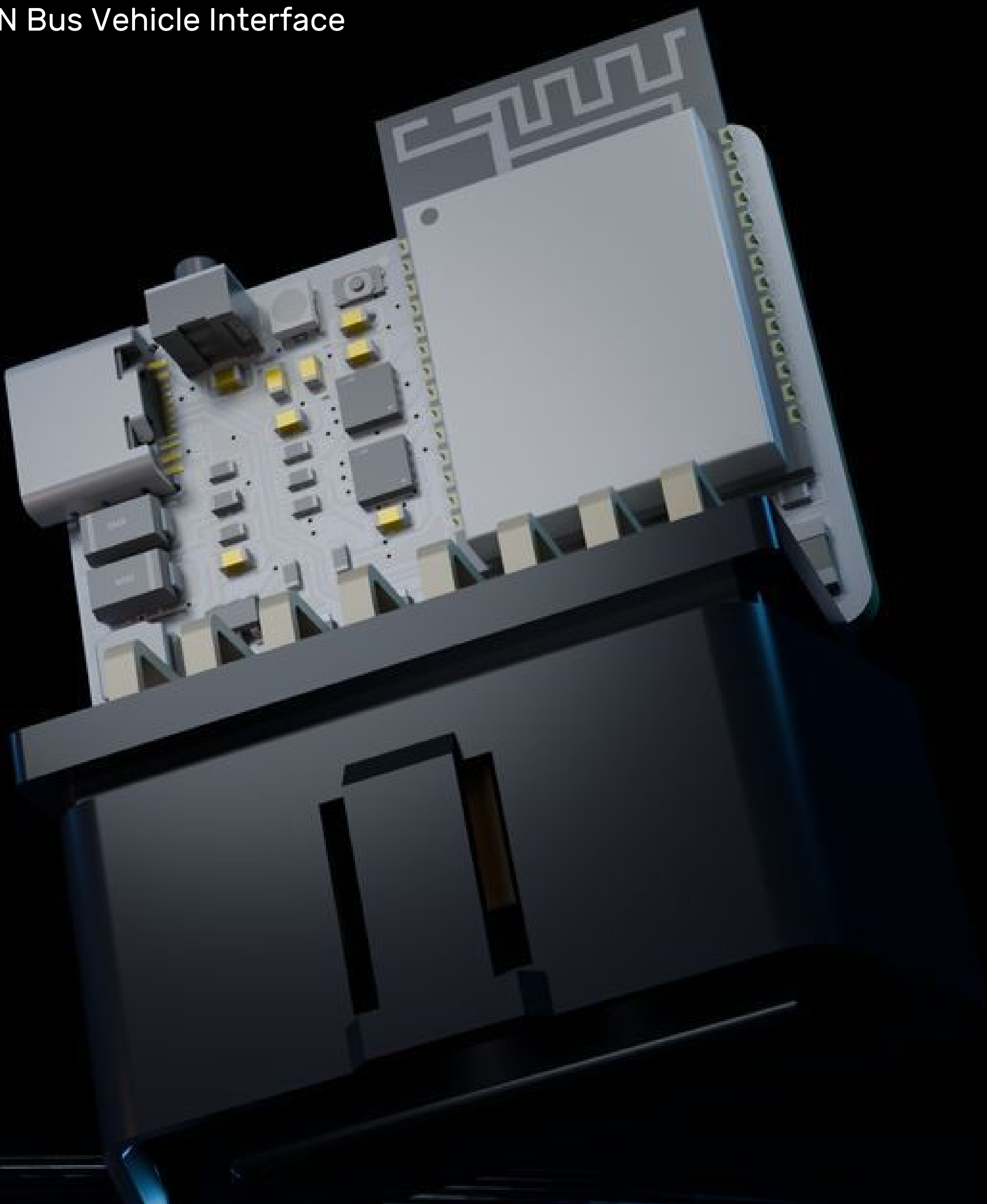
CANOpener SE

Dual CAN Bus Vehicle Interface



Hardware Documentation

Standard Edition

Hardware Revision A

Introduction

CANOpener SE is a compact dual CAN bus interface designed to connect directly to vehicle networks through the OBD-II diagnostic port.

At the heart of CANOpener SE is an Espressif ESP32-C5 microcontroller with two independent CAN-FD controllers and integrated wireless connectivity. Two Texas Instruments TCAN3414 CAN transceivers provide the physical interface between the controller and the vehicle CAN networks.

The device connects directly to two CAN buses available through the OBD-II connector and supports both Classical CAN and CAN FD communication. CANOpener SE can be powered directly from the vehicle or through USB-C for development and configuration.

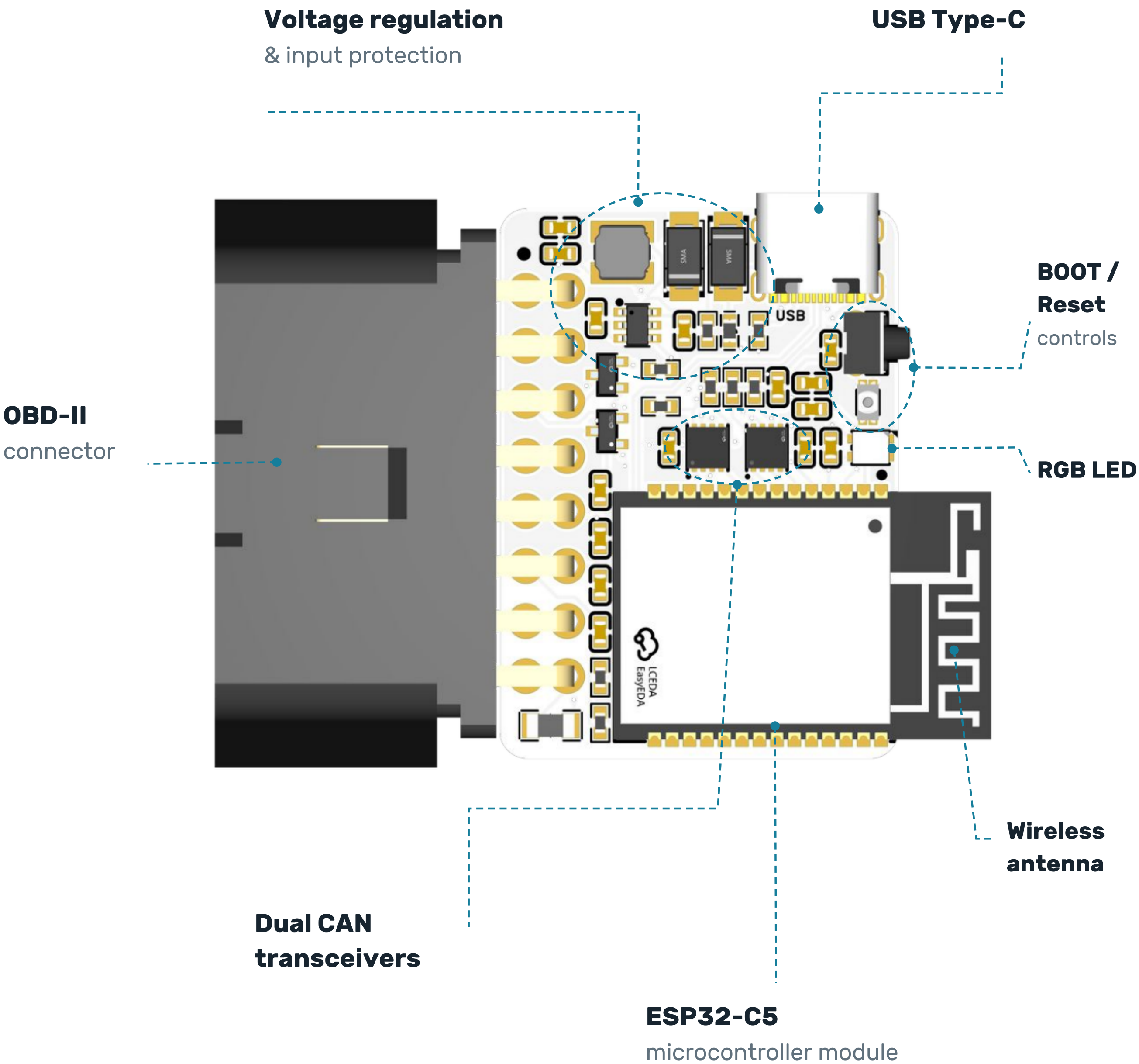
Onboard voltage regulation, input protection, CAN bus protection, an addressable RGB status LED, and dedicated Boot and Reset controls are integrated into the board.

Electrical specifications

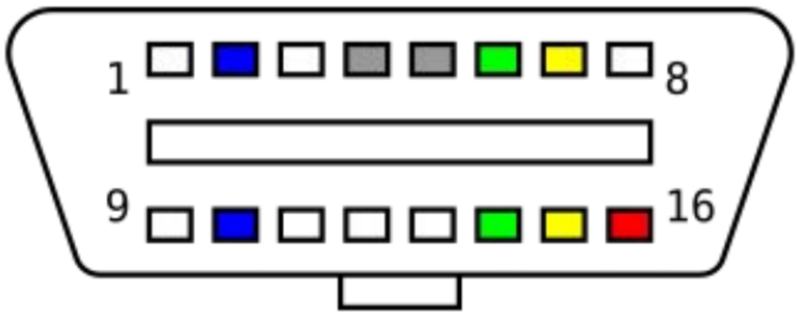
Parameter	Specification
Vehicle input voltage	Up to 24 V
Typical input voltage	12 V
USB input voltage	5 V
Logic / I/O voltage	3.3 V
Operating current	Not specified
CAN interfaces	2
CAN protocols	Classical CAN, CAN FD
Maximum CAN data rate	5 Mbps

Hardware overview

Board features at a glance



OBD-II connector



Connector pinout

Pin	Connection	Pin	Connection
1	NC	9	NC
2	NC	10	NC
3	CAN 2 High (CAN1)	11	CAN 2 Low (CAN1)
4	GND	12	NC
5	NC	13	NC
6	CAN 1 High (CAN0)	14	CAN 1 Low (CAN0)
7	NC	15	NC
8	NC	16	Vehicle Power

GPIO reference

ESP32-C5-WROOM-1-N16R8 pin assignments for the BOOT button, addressable RGB LED, dual CAN interfaces, and CAN standby control.

CAN 1 corresponds to schematic channel CAN_0; CAN 2 corresponds to CAN_1. GPIO numbers below identify MCU signals, not module pad numbers.

Control and CAN signals

Function	Schematic net	GPIO	Module pad
BOOT button	BOOT	28	15
RGB LED data	RGB_LED	2	4
CAN 1 RX	CAN_0_RXD	0	6
CAN 1 TX	CAN_0_TXD	1	7
CAN 2 RX	CAN_1_RXD	7	9
CAN 2 TX	CAN_1_TXD	8	10
CAN standby (STB)	CAN_STB	3	5

Signal notes

RX and TX are named from the microcontroller perspective: RX receives data from the CAN transceiver; TX sends data to the CAN transceiver.

The RGB LED uses one data GPIO. The schematic shows a single CAN_STB control signal on GPIO3.

Physical specifications